ICT2152 · Object Oriented Analysis and Design

Practical Tutorial: Refine a Design

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Start with a use case, draw a domain model, identify service operations, and sketch the sequence of calls. Check that every object has a clear purpose and that role checks occur before sensitive data access.

2. Example and reasoning

Design a material publishing flow. Include a missing-file alternate path, an unauthorized lecturer path, and a successful path. Specify where file validation occurs and how the service prevents publishing invalid references.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “practical tutorial: refine a design” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.